Band-1 Hadamard Order 2708 — Miyamoto (H_676 Paley-II) + Paley-I Cross-Check; Gap 2052 Still OPEN

H_676/C_678/H_2708 PASS (Miyamoto + Paley-I); 2708 in SPK via Paley I q=2707; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_2692_Miyamoto`; `2026-09-15_Band1_Hadamard_2564_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_2708_miyamoto/`.
Cutoff. 2026-09-15 (~02:28Z / ~10:28PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 676 and 2708 only** (verified $\max|HH^\top - nI| = 0$), via **two independent paths** for order 2708 (Miyamoto primary; Paley type I cross-check). It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. Order 2708 **enters SPK** via Paley I ($q = 2707$); it is **not** a Miyamoto-only bonus (contrast prior 2692/2564/2276/2228/2164). Honest SPK audit: Paley-II **FAIL** ($q = 1353$ not prime); Kronecker **FAIL** ($2708 = 4 \cdot 677$, 677 not Hadamard-admissible); Paley-I **PASS**. `clay_closed = false`.

Abstract

Prior Band-1 ships constructed Miyamoto-eligible orders through 2692 on box (including Miyamoto-only bonuses 2564 and 2692). This increment constructs the **next preferred** Miyamoto-eligible target after $q = 673$: **order 2708** ($2708 = 4 \cdot 677$, 677 prime $\equiv 1 \pmod 4$). Seed H_{676} via Paley type II ($q = 337$ prime $\equiv 1 \pmod 4$; order $2(q+1) = 676$). Symmetric Paley conference C_{678} over prime 677. $H_{2708} = \text{Miyamoto}(H_{676}, C_{678})$. Independently, H_{2708} also verifies via **Paley type I** with $q = 2707$ (explains SPK membership). Unit tests **26/26 PASS**. SPK next gap remains **2052** (OPEN). The **full Hadamard conjecture remains OPEN**.

Keywords. Hadamard matrix; order 2708; Miyamoto; Paley type II; Paley type I; Paley conference; constructive gap 2052; Band-1; non-claim.

1. Statement lock

Definition. A Hadamard matrix of order n is an $n \times n$ matrix H with entries in $\{\pm 1\}$ satisfying $HH^\top = nI$.

Finite lemmas (this increment). 1. H_{676} constructed and verified (Paley type II, $q = 337$). 2. C_{678} constructed and verified (Paley conference, $q = 677$): $CC^\top = 677I$, zero diagonal. 3. H_{2708} constructed and verified (Miyamoto). **Also** constructed and verified via Paley type I ($q = 2707$). **Not** the Hadamard conjecture. 4. SPK: $2708 \in$ SPK (via Paley I). Paley-II and Kronecker paths for 2708 **rejected**. Next constructive gap remains **2052** (OPEN).

2. Arithmetic and eligibility

Quantity	Value
Target order	$2708 = 4 \cdot 677$
$q = 677$	prime, $677 \equiv 1 \pmod 4 \rightarrow$ Miyamoto-eligible
Needed H_{q-1}	H_{676}
Needed conference	order $678 = q + 1$
H_{676} path (used)	Paley type II, $q = 337$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 676$
Conference path	Symmetric Paley over prime 677
SPK path for 2708	Paley type I, $q = 2707$ prime $\equiv 3 \pmod 4$; order $q + 1 = 2708$
Paley-II for 2708	**REJECTED**: $q' = 1353$ not prime
Kronecker for 2708	**REJECTED**: only 4×677; $677 \not\equiv 0 \pmod 4$ (not admissible)

3. Construction results

Object	Method	$\max \lvert HH^\top - nI \rvert$ (or conf)	Wall (s)	Status
H_{676}	Paley type II, $q = 337$	0	0.2231	**PASS**
C_{678}	Paley conference, $q = 677$	0	0.4037	**PASS**
H_{2708}	Miyamoto(H_{676}, C_{678})	0	8.5842	**PASS**
H_{2708} P-I	Paley type I, $q = 2707$	0	12.1929	**PASS**

Primary Miyamoto pipeline wall (incl. unit tests + SPK refresh + Paley-I cross-check) $\approx$ 22.3416 s. Compressed ± 1 matrices saved as `H676_pm1.npz`, `C678_pm1.npz`, `H2708_pm1.npz`, `H2708_paleyI_pm1.npz` (int8).

4. SPK closure note

Seeds: all admissible ≤ 2000 $\cup$ on-box prior ships $\cup$ verified H_{676} $\cup$ Sylvester $\cup$ Paley I/II, closed under Kronecker (limit 4000). **Miyamoto is not folded into SPK**; order 2708 enters SPK **because** Paley I applies ($q = 2707$).

Quantity	Value
Known count	882
2708 in SPK	**True** (Paley I $q = 2707$)
676 in SPK	True (Paley II $q = 337$)
2692 in SPK	False (prior Miyamoto bonus)
2564 in SPK	False (prior Miyamoto bonus)
Next standard gap	**2052** (OPEN)
Missing first (sample)	2052, 2060, 2076, 2092, 2108, 2148, 2156, 2172, 2212, 2284…

5. Gap 2052 (still OPEN; non-finding ONLY)

This ship does **not** fill gap 2052. Prior triage stands: Paley-I/II / Kronecker / Miyamoto-Paley-conf all **FAIL** for order 2052. Claim level for 2052 remains method triage / non-finding ONLY. Non-finding $\neq$ counterexample; Hadamard conjecture remains OPEN. No H_{2052} matrix is claimed.

6. Unit tests (26/26)

Test	Result
`H4_sylvester`	PASS
`H20_paley_I_control`	PASS
`paley_II_q5_H12`	PASS

Test	Result
paley_II_q13_H28	PASS
miyamoto_H20_from_H4_conf6	PASS
paley_I_q11_H12	PASS
q337_is_prime	PASS
q337_equiv_1_mod_4	PASS
H676_paley_II_order_formula	PASS
q677_is_prime	PASS
q677_equiv_1_mod_4	PASS
n2708_equals_4_times_677	PASS
H676_order_formula	PASS
C678_order_formula	PASS
paley_I_H2708_path_eligible	PASS
paley_II_H2708_path_rejected	PASS
kronecker_H2708_path_rejected	PASS
q2707_is_prime_equiv_3_mod_4	PASS
paley_I_order_2708_formula	PASS
reject_all_ones_H8	PASS
conference_6_paley_control	PASS
miyamoto_eligible_2708_arith	PASS
conference_678_shape_expected	PASS
gap_2052_not_filled_by_this_ship	PASS
2708_in_spk_via_paley_I_arith	PASS
paley_II_seed_pattern_control	PASS

7. Next optional targets

Miyamoto-eligible primes $q > 677$, $q \equiv 1 \pmod 4$ (sample):

q	$n = 4q$	Need H_{q-1}	On-box path notes
701	2804	700	PaleyII(q=349)
709	2836	708	PaleyII(q=353)
733	2932	732	—
757	3028	756	—
761	3044	760	Double(PaleyI half=380), Kronecker(380x2)
769	3076	768	Double(PaleyI half=384), Kronecker(4x192)
773	3092	772	—
797	3188	796	PaleyII(q=397)

Preferred next. Method-attack remaining SPK gap **2052** (GS/SDS for order ~513 if literature

seeds appear), **or** construct Miyamoto $q = 701 \to n = 2804$ (seed Paley-II $q = 349$), **or** Track B privacy / Object 1 meter await. GATE-04 remains LOCKED.

8. Non-claims (explicit)

- Not the full Hadamard conjecture.
- Not Clay-closed / finished-ToE / RH / P≠NP / YM / Frankl / NS-solved / AC-solved.
- Gap **2052** remains **OPEN**.
- Order 2708 is **in SPK** (Paley I); not a Miyamoto-only bonus.
- GATE-04 collection **LOCKED**; no invented J.
- `clay_closed = false`.

9. Artifacts

Lab: `/workspace/experiments/band1/hadamard_2708_miyamoto/`
Verifier: `verify_hadamard_2708.py`
Results: `results.json`, `SUMMARY.txt`, `literature_ledger.json`
Matrices: `H676_pm1.npz`, `C678_pm1.npz`, `H2708_pm1.npz`, `H2708_paleyI_pm1.npz`

ZoraASI Band-1 · finite constructions only · honesty gates held.